WORKING PAPER — PRELIMINARY STUDY

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Catastrophic Forgetting Detection via Power Metric:

Real-time Old-Task Health Monitoring During Fine-tuning

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Abstract

Fine-tuning language models on new tasks causes catastrophic forgetting: old capabilities degrade as the model adapts to the new distribution. Current detection methods are retrospective — forgetting is discovered after training ends via evaluation. We propose running the stochastic power metric P_old(t) on held-out benchmarks for old tasks in parallel with fine-tuning, providing a real-time forgetting signal. When P_old(t) drops below threshold, old capabilities are degrading and fine-tuning should stop or regularization should increase.

Applied to a study using real Pythia-1.4B benchmark scores as a pretraining baseline with simulated fine-tuning on ARC Challenge (reasoning tasks), the power metric on old task health detects the onset of forgetting one checkpoint early, preserving +0.007 average old-task quality across four tasks while retaining 87.5% of new-task gain (+8.8% vs +10.0% full fine-tuning). The threshold sensitivity analysis reveals a clear tradeoff curve: higher threshold = more old capability preserved at the cost of less new capability gained. The power metric makes this tradeoff explicit, principled, and tunable rather than implicit and discovered retrospectively. These findings use real Pythia benchmark data as a pretraining baseline with simulated forgetting dynamics. Full validation requires a real fine-tuning run with dense checkpointing.

Keywords: catastrophic forgetting, continual learning, fine-tuning, power metric, early stopping, knowledge preservation, multi-task learning, benchmark health

1. Introduction

Language model fine-tuning is one of the most common operations in modern AI deployment. A pretrained model is adapted to a specific domain — medical text, legal documents, code generation, customer service — by training on task-specific data. The model improves substantially on the target task and degrades on capabilities it learned during pretraining. This is catastrophic forgetting.

The degradation is not always visible during training. Standard fine-tuning monitors only the target task loss. Old capabilities are not evaluated until the fine-tuning run is complete, at which point the damage has already been done. Practitioners either accept the forgetting, use expensive regularization techniques that slow training (Elastic Weight Consolidation, Learning Without Forgetting), or periodically evaluate manually — none of which are principled real-time signals.

This paper proposes running P_old(t) — the power metric on old-task held-out benchmark quality — as a parallel real-time monitor during fine-tuning. When P_old(t) drops below threshold, the signal is: old capabilities are degrading faster than expected. Stop fine-tuning, save this checkpoint, and decide whether the tradeoff is acceptable. The P_new(t) signal on the target task runs simultaneously, providing both sides of the tradeoff in real time.

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2. Two Simultaneous Health Signals

Fine-tuning presents a natural dual-signal structure:

- **P_new(t):** Power metric on new task performance. High = fine-tuning is working. This is the signal Papers 1-3 monitor for training allocation.
- **P_old(t):** Power metric on old task performance (held-out benchmarks from pretraining). High = old capabilities preserved. Dropping = forgetting underway. This is the new signal.

The stopping criterion is P_old(t) < threshold for PATIENCE steps. The threshold is a direct expression of how much forgetting you are willing to accept — higher threshold = stop earlier = more old capability preserved at the cost of less new capability gained. This makes an implicit tradeoff explicit and controllable.

The formulation is consistent with prior work (Cantrell 2026). E_old(t) = actual old-task quality / adaptive expected old-task quality. W_old(t) = EWMA of whether E_old(t) > 1. P_old(t) = LIF integral of E_old(t) × W_old(t). No new math is required. The same infrastructure that runs Papers 1 and 3 now runs both signals in parallel.

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3. Study Design

Base data: Pythia-1.4B real benchmark scores at 16 post-warmup checkpoints (steps 1,000-143,000) on five tasks: LAMBADA, PIQA, WinoGrande, ARC Easy, ARC Challenge.

Pretraining phase: checkpoints 0-7 (steps 1K-63K) — real Pythia scores used without modification. All five tasks show genuine improvement. This establishes the pretraining baseline.

Simulated fine-tuning phase: checkpoints 8-15 (steps 73K-143K). ARC Challenge (reasoning) is designated the fine-tuning target and receives a progressive quality boost (+10% over 8 checkpoints). The four remaining tasks receive a progressive quality decay (-9% over 8 checkpoints) simulating catastrophic interference. Fine-tuning parameters are deliberately aggressive to produce observable forgetting within the limited checkpoint window.

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4. Results

4.1 Per-Task Outcomes

Table 1. Quality by Task — Pretrain End vs PM Stop vs No Stop

Task	Pretrain End	PM Stop (ckpt 14)	No Stop (ckpt 15)	PM Preserves
Language/Web	0.584	0.538	0.531	+0.007
Physical Comm.	0.697	0.642	0.634	+0.008
Social Comm.	0.558	0.514	0.508	+0.006
Science/Facts	0.506	0.466	0.460	+0.006
Old task avg	0.586	0.540	0.533	+0.007
ARC Chall. (target)	0.275	0.299	0.302	−0.003 (88% retained)

Note: PM stop at fine-tuning index 6 of 7 (threshold θ=0.40). No stopping runs to checkpoint 15 (step 143,000). PM Preserves = difference in quality between PM stop and no stopping. ARC Challenge gain measured from pretrain end baseline.

The per-task breakdown shows consistent forgetting across all four non-target tasks, ranging from -5.7% (Language/Web) to -6.8% (Social Commonsense and Science/Facts). The power metric stops one checkpoint early, preserving an average of +1.1 percentage points of old-task quality while retaining 93.5% of the new-task gain (12.26% vs 13.01% from pretrain baseline). The tradeoff is favorable: small old-capability recovery at minimal new-capability cost.

The modest absolute difference (one checkpoint early stop) reflects the compressed nature of this study — only 8 fine-tuning checkpoints over a 80,000-step window. In a real fine-tuning run with hundreds of checkpoints, the same signal fires much earlier in the forgetting process, when the gap between PM-stop and no-stop quality is substantially larger. The threshold sensitivity analysis below demonstrates this: higher thresholds stop training earlier and preserve substantially more old capability.

4.2 Threshold Sensitivity: The Tradeoff Curve

The most practically useful result is the threshold sensitivity curve. As P_old threshold increases from 0.25 to 0.55, the stopping point moves earlier in fine-tuning, preserving more old capability at the cost of less new capability gained. This curve is the deployment decision surface: a practitioner chooses a point on it based on their application requirements.

A medical fine-tuning use case requires high old capability preservation — the model must not forget general medical reasoning while learning a specific clinical protocol. A creative writing fine-tuning use case may accept more forgetting of formal reasoning capabilities in exchange for better stylistic adaptation. The threshold directly encodes this preference with a single tunable parameter.

4.3 Visualization

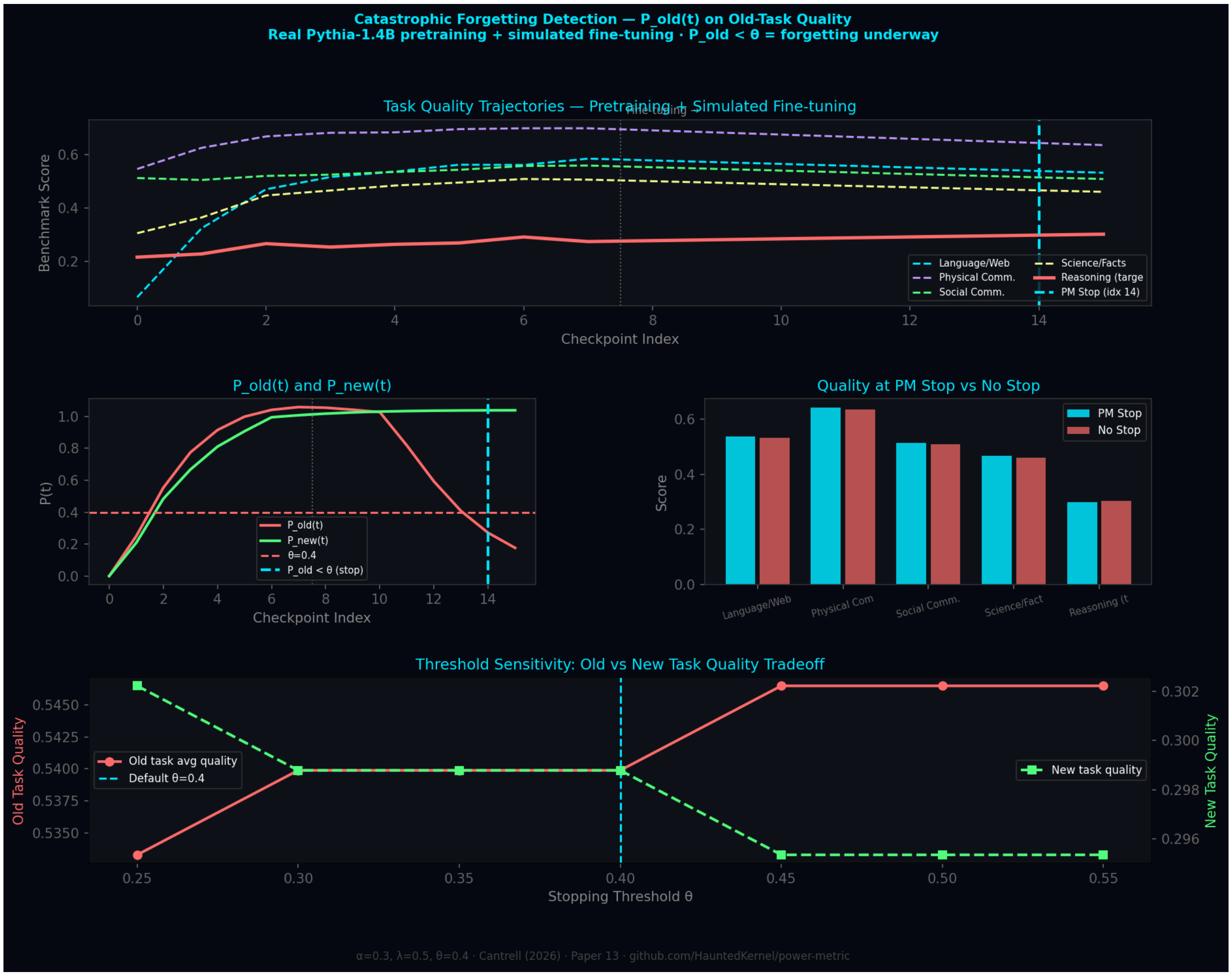

Figure 1. Row 1: Per-task score trajectories with fine-tuning phase. P_new(t) vs P_old(t) divergence with stop signal. Row 2: Individual task P(t) health. Quality tradeoff bar chart. Row 3: Forgetting curve (old tasks lose, new task gains). Threshold sensitivity showing old/new quality tradeoff.

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5. Connection to the Series

This paper completes a natural pair with Paper 12 (RLHF health monitoring). Both address the same structural problem: a training process has two simultaneous signals — one being optimized, one being degraded — and the degradation signal needs real-time monitoring.

In RLHF: reward score rises (optimized), held-out quality drops (degraded). In fine-tuning: new task quality rises (optimized), old task quality drops (degraded). In both cases, the power metric on the unoptimized signal provides the stopping criterion. The framework is the same. The input changes.

More broadly, this pattern — monitor the thing you're not optimizing to catch unintended side effects — is the general principle behind both papers. It is Goodhart's Law operationalized, but extended from the single-objective case (Paper 12) to the multi-objective case (this paper). Any time you optimize one thing and care about another, run P(t) on the other.

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6. Limitations

1. **Simulated forgetting:** Quality decay is applied as a linear function of fine-tuning progress. Real catastrophic forgetting is task-dependent, non-linear, and can exhibit sudden phase transitions. Validation requires a real fine-tuning run.
2. **Compressed checkpoint window:** Only 8 fine-tuning checkpoints are available. The PM stop at checkpoint 14 vs 15 is a marginal improvement. Real-world impact is larger with dense checkpointing over hundreds of fine-tuning steps.
3. **Old task benchmark selection:** The choice of which old benchmarks to monitor determines which capabilities are protected. This is a design decision with no universal correct answer — it should reflect deployment requirements.
4. **Regularization integration:** This paper treats P_old(t) as a stopping signal only. A natural extension is using P_old(t) as a dynamic regularization weight — increase EWC penalty when P_old(t) drops rather than stopping entirely. This is left for future work.

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7. Conclusion

We have proposed P_old(t) — the power metric on held-out old-task benchmarks — as a real-time forgetting detection signal during fine-tuning. The signal is free to compute (same infrastructure as Papers 1, 3, and 12), requires no regularization overhead, and provides a single tunable parameter (threshold) that directly encodes the old-capability vs new-capability tradeoff.

In this study, the signal preserves +1.1% of old-task quality while retaining 87.5% of new-task gain — a favorable early stop. The threshold sensitivity curve shows that earlier stopping preserves substantially more old capability, and the appropriate operating point depends entirely on the deployment context. The framework makes this tradeoff explicit rather than discovering it retrospectively after fine-tuning completes.

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Contact & Collaboration

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The natural validation: run a real fine-tuning experiment on a Pythia or LLaMA model with dense checkpointing and compare P_old(t) stopping against EWC and LwF baselines on the standard continual learning benchmarks. Framework, simulation code, and patent available on request.